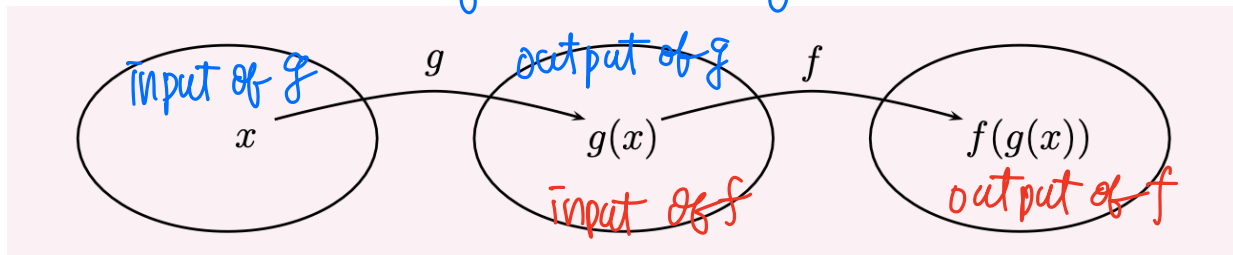


$$D_{\neq} = \{x \mid x \in (-\infty, -2) \cup (-2, \infty)\} = \{x \mid x \in \mathbb{R}, \text{ but } x \neq -2\}$$

3. Complete the definition of the Composition of Functions:

Let $f(x)$ and $g(x)$ be two functions. The composition of the function f with g is denoted by

$(f \circ g)(x)$ and is defined by the equation $(f \circ g)(x) := f(g(x))$.



The domain of the composition of the function $f \circ g$ is the set of all x such that x is the input of $g(x)$ and output of $g(x)$ is the domain of $f(x)$.

The notation of the domain of the composition of the function $f \circ g$ is

$D_{f \circ g} = \{x \mid x \text{ is from } D_g \text{ and output of } g(x) \text{ is from } D_f\}$

4. Are $f(g(x))$ and $g(f(x))$ the same functions?

$f(p) = p \cdot 0.8$ $g(p) = p - 500$ $g(f(p)) = f(p) - 500 = p \cdot 0.8 - 500$

$f(g(p)) = g(p) \cdot 0.8 = (p - 500) \cdot 0.8$

5. Find $(f \circ g)(x)$ for the following functions and state their domains.

a) $f(x) = x^2 + 2$ and $g(x) = x - 3$

$f(g(x)) = f(x - 3) = (x - 3)^2 + 2 = x^2 - 6x + 9 + 2 = x^2 - 6x + 11$

$D_{f \circ g} = (-\infty, \infty) = \{x \mid x \in \mathbb{R}\}$

b) $f(x) = \frac{2}{x-3}$ and $g(x) = x^2 + 2x$

$f(g(x)) = \frac{2}{g(x) - 3} = \frac{2}{x^2 + 2x - 3}$

$x^2 + 2x - 3 = (x - 1)(x + 3) = 0$

$x - 1 = 0$ or $x + 3 = 0$

$x = 1$, or $x = -3$

$D_{f \circ g} = \{x \mid x \in \mathbb{R}, \text{ but } x \neq 1 \text{ and } x \neq -3\}$